

## Session 8

### RTM Assignment – Requirements Traceability Matrix

1.

**Pick any one feature from a popular app you use daily (for example: Instagram signup, Flipkart add-to-cart, or Paytm recharge) and write 5 clear requirements for that feature.**

I will use the **Instagram Signup feature** as an example.

A requirement tells us **what the application should do**. For Instagram signup, the main purpose is to allow a new user to create an account successfully.

| Requirement ID | Requirement Description                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|
| R1             | The user should be able to enter a valid mobile number or email address.                                                           |
| R2             | The application should allow the user to create a password that satisfies the required password rules.                             |
| R3             | The application should send an OTP/verification code to verify the user's mobile number or email.                                  |
| R4             | The Signup button should allow registration only when all required information is entered correctly.                               |
| R5             | After successful verification, the application should create the user's account and show the appropriate confirmation/home screen. |

**In simple words:** These five requirements describe what we expect the signup feature to do. During testing, we create test cases to check whether each requirement is working correctly.

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**2. For the requirements you wrote in Task 1, create a table mapping each requirement to at least one test**

case that would verify it works as intended.<br><br><em><strong>Hint:</strong> Your table should have columns: Requirement ID, Requirement Description, Test Case ID, Test Case Description.</em>

### Map each requirement to a test case

A **test case** is a specific action or scenario used by a tester to verify a requirement.

For example, if the requirement says "*The user should be able to enter a valid email*," the test case can check whether a valid email is accepted.

| Requirement ID | Requirement Description                                               | Test Case ID | Test Case Description                                                                           |
|----------------|-----------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------|
| R1             | User should be able to enter a valid mobile number or email.          | TC01         | Enter a valid email address and verify that it is accepted.                                     |
| R2             | User should be able to create a password according to password rules. | TC02         | Enter a password that satisfies all password requirements and verify it is accepted.            |
| R3             | Application should send an OTP/verification code.                     | TC03         | Enter a valid mobile/email and verify that the OTP is sent.                                     |
| R4             | Signup button should work only when required information is valid.    | TC04         | Enter all valid required details and verify that the Signup button becomes enabled.             |
| R5             | Application should create the account after successful verification.  | TC05         | Complete signup with valid details and OTP and verify that the account is created successfully. |

Easy way to remember:

Requirement = What should happen

Test Case = How we check that it happens

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**3. Extend your table from Task 2 by adding a 'Defect ID' column. Imagine that during testing, you find 2**

**bugs related to your test cases. Link these defects to the relevant test cases in your table.**  
**Hint: Defects can be things like 'Signup button not enabled' or 'OTP not received'.**

### Add defects to the table

A **defect** is a problem found during testing where the actual result is different from the expected result.

Suppose we find two bugs:

- **DEF01:** OTP is not received by the user.
- **DEF02:** Signup button remains disabled even after all valid information is entered.

We can add these defects to our table.

| Requirement ID | Requirement Description                                              | Test Case ID | Test Case Description                                                               | Defect ID    |
|----------------|----------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|--------------|
| R1             | User should be able to enter a valid mobile number or email.         | TC01         | Enter a valid email and verify that it is accepted.                                 | —            |
| R2             | User should be able to create a valid password.                      | TC02         | Enter a valid password and verify that it is accepted.                              | —            |
| R3             | Application should send an OTP/verification code.                    | TC03         | Enter a valid mobile/email and verify that the OTP is sent.                         | <b>DEF01</b> |
| R4             | Signup button should work when required information is valid.        | TC04         | Enter all valid required details and verify that the Signup button becomes enabled. | <b>DEF02</b> |
| R5             | Application should create the account after successful verification. | TC05         | Complete signup with valid details and OTP and verify that the account is created.  | —            |

Here, **DEF01 is linked to TC03** because the OTP test failed. Similarly, **DEF02 is linked to TC04** because the Signup button did not behave as expected.

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**4. Explain in 3-4 sentences the difference between forward traceability and backward traceability in the context of your table. Give one example of each using your requirements and test cases.**

## **Forward Traceability and Backward Traceability**

**Forward traceability** means tracing a requirement **forward to its test case**. It helps us check whether every requirement has been tested. For example, requirement **R3**, which says that an OTP should be sent, is linked forward to **TC03**, which tests whether the OTP is received.

**Backward traceability** means tracing a test case **back to the requirement** from which it was created. For example, **TC04** tests the Signup button, so we can trace it backward to **R4**, which specifies that the Signup button should work when valid information is entered.

In simple words, **Forward = Requirement → Test Case**, while **Backward = Test Case → Requirement**. When both directions are maintained, it is called **bi-directional traceability**.

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**5. Create a bi-directional RTM (Requirements Traceability Matrix) for the login feature of a food delivery app like Zomato, showing how each requirement is linked to test cases and how test cases trace back to requirements.**  
**Constraint:** Include at least 3 requirements and 5 test cases in your matrix.

# Bi-directional RTM for Zomato Login

Now let's create an RTM for the **Login feature of a food delivery app like Zomato**.

The purpose of the login feature is to allow registered users to securely access their accounts.

## Requirements

- **R1:** User should be able to enter a registered mobile number/email.
- **R2:** The application should validate the user's login credentials/OTP.
- **R3:** User should be successfully logged in after entering valid details.
- **R4:** The application should display an appropriate error message for invalid details.

## Test Cases

- **TC01:** Enter a registered mobile number and verify that it is accepted.
- **TC02:** Enter an unregistered mobile number and verify that an appropriate error is displayed.
- **TC03:** Enter the correct OTP and verify that it is accepted.
- **TC04:** Enter an incorrect OTP and verify that login is rejected with an error message.
- **TC05:** Enter valid login details and verify that the user is successfully logged in.

## Bi-directional RTM

| Requirement ID | Requirement Description                                        | Test Case ID | Test Case Description                                                 | Traceability |
|----------------|----------------------------------------------------------------|--------------|-----------------------------------------------------------------------|--------------|
| R1             | User should be able to enter a registered mobile number/email. | TC01         | Enter a registered mobile number and verify that it is accepted.      | R1 → TC01    |
| R1             | User should be able to enter a registered mobile number/email. | TC02         | Enter an unregistered mobile number and verify the appropriate error. | R1 → TC02    |
| R2             | Application should validate the login OTP.                     | TC03         | Enter the correct OTP and verify that it is accepted.                 | R2 → TC03    |
| R2             | Application should validate the login OTP.                     | TC04         | Enter an incorrect OTP and verify that login is rejected.             | R2 → TC04    |

|    |                                                               |      |                                                                   |           |
|----|---------------------------------------------------------------|------|-------------------------------------------------------------------|-----------|
| R3 | User should successfully log in after entering valid details. | TC05 | Enter valid login details and verify successful login.            | R3 → TC05 |
| R4 | Application should display an error for invalid details.      | TC02 | Enter an unregistered mobile number and verify the error message. | R4 → TC02 |
| R4 | Application should display an error for invalid details.      | TC04 | Enter an incorrect OTP and verify the error message.              | R4 → TC04 |

### How this is bi-directional

You can read the table in **both directions**:

#### Forward:

R1 → TC01, TC02  
 R2 → TC03, TC04  
 R3 → TC05  
 R4 → TC02, TC04

This tells us which test cases are used to test each requirement.

#### Backward:

TC01 → R1  
 TC02 → R1, R4  
 TC03 → R2  
 TC04 → R2, R4  
 TC05 → R3

This tells us which requirement each test case is testing.